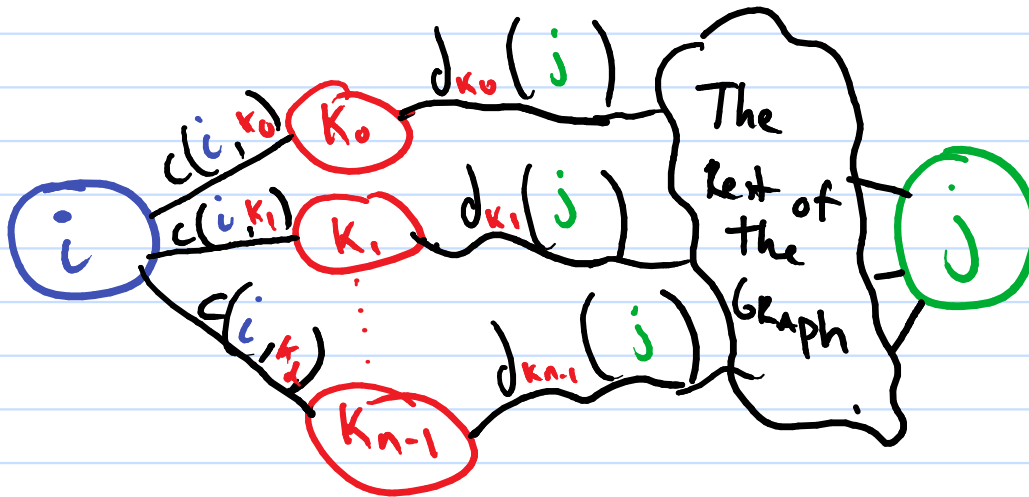


Bellman-Ford Equation

$$d_i(j) = \min_k \{ c(i, k) + d_k(j) \}$$

Annotations for the equation:

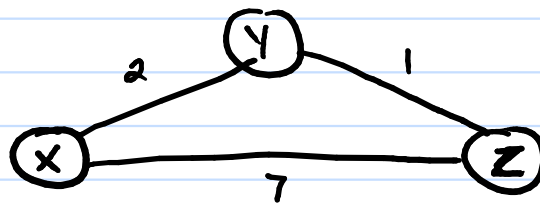
- $d_i(j)$: shortest path from i to j
- $\min_k$: minimum over all nodes k , where k is a neighbor to i
- $c(i, k)$: edge cost for (i, k)
- $d_k(j)$: shortest path from k to j



1. The path from i to j must start by going through one of i 's neighbors, k .
2. The cost of that path is $c(i, k) + d_k(j)$
3. Which ever of these paths is shortest is the shortest path from i to j and that cost is $d_i(j)$

* The algorithm solves the B-F equation in a distributed, asynchronous manner.

Figure 5.6



Local Dist. Vectors

D_i for $i \in \{X, Y, Z\}$

STEP 0

X	x	y	z	Y	x	y	z	Z	x	y	z
x	0	2	7	x	∞	∞	∞	x	∞	∞	∞
y	∞	∞	∞	y	2	0	1	y	∞	∞	∞
z	∞	∞	∞	z	∞	∞	∞	z	7	1	0

STEP 1a GET vector from neighbors

X	x	y	z	Y	x	y	z	Z	x	y	z
x	0	2	7	x	0	2	7	x	0	2	7
y	2	0	1	y	2	0	1	y	2	0	1
z	7	1	0	z	7	1	0	z	7	1	0

Step 1b recompute "personal" Distance Vector

$$EQ: D_i(j) = \min_k \{ \underbrace{c(i, k)}_{\substack{\text{EDGE COST} \\ \text{FROM } i \text{ TO } k}} + D_k(j) \}$$

$D_{\text{row}}(\text{col})$
Entry in the Dist. Vector Matrix

Example for Node y

$$D_Y(Y) = 0$$

$$D_Y(Z) = \min \{ c(Y, X) + D_X(Z), c(Y, Z) + D_Z(Z) \}$$

$$= \min \{ 2 + 7, 1 + 0 \}$$

$$= \min \{ 9, 1 \} = 1$$

$$D_Y(X) = \min \{ c(Y, X) + D_X(X), c(Y, Z) + D_Z(X) \}$$

$$= \min \{ 2 + 0, 1 + 7 \}$$

$$= \min \{ 2, 8 \} = 2$$

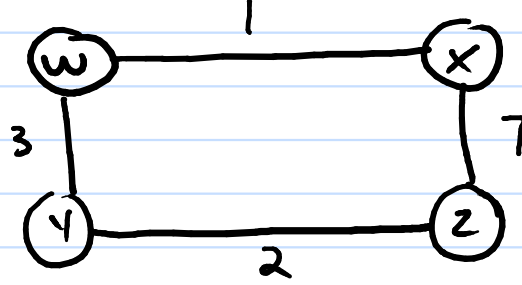
For Each Node in the Graph

For Each neighbor of y

→ Repeat for each Node using that nodes "Matrix"

Ex 2

Not fully connected graph



Optional Paths

$$\begin{aligned} d_w(y) &= d_y(w) = 3 \\ d_w(x) &= d_x(w) = 1 \\ d_w(z) &= d_z(w) = 5, y \\ d_x(z) &= d_z(x) = 6, w \\ d_x(y) &= d_y(x) = 4, w \\ d_z(y) &= d_y(z) = 2 \end{aligned}$$

$$i \quad j \quad \begin{matrix} & w & x & y & z \\ \begin{matrix} w \\ x \\ y \\ z \end{matrix} & \begin{bmatrix} 0 & 1 & 3 & \infty \\ 1 & 0 & \infty & 7 \\ 3 & \infty & 0 & 2 \\ \infty & 7 & 2 & 0 \end{bmatrix} \end{matrix}$$

Adjacency Matrix $\Rightarrow c(i, j)$
Giving Path costs

TARGET D.Vect. SET

$$\begin{aligned} D_w &= [0, 1, 3, \infty] \\ D_x &= [1, 0, \infty, 7] \\ D_y &= [3, \infty, 0, 2] \\ D_z &= [\infty, 7, 2, 0] \end{aligned}$$

Initial Distance Vectors for each Node.

$$\begin{matrix} & w & x & y & z \\ \begin{matrix} w \\ x \\ y \\ z \end{matrix} & \begin{bmatrix} 0 & 1 & 3 & 5 \\ 1 & 0 & 4 & 6 \\ 3 & 4 & 0 & 2 \\ 5 & 6 & 2 & 0 \end{bmatrix} \end{matrix}$$

One Step For z:

init D

$$\begin{bmatrix} \infty & \infty & \infty & \infty \\ \infty & \infty & \infty & \infty \\ \infty & \infty & \infty & \infty \\ \infty & 7 & 2 & 0 \end{bmatrix}$$

Get D_i from neighbors x, y

$$\begin{matrix} & w & x & y & z \\ \begin{matrix} w \\ x \\ y \\ z \end{matrix} & \begin{bmatrix} \infty & \infty & \infty & \infty \\ 1 & 0 & \infty & 7 \\ 3 & \infty & 0 & 2 \\ \infty & 7 & 2 & 0 \end{bmatrix} \end{matrix}$$

C

$$\begin{matrix} & w & x & y & z \\ \begin{matrix} w \\ x \\ y \\ z \end{matrix} & \begin{bmatrix} 0 & 1 & 3 & \infty \\ 1 & 0 & \infty & 7 \\ 3 & \infty & 0 & 2 \\ \infty & 7 & 2 & 0 \end{bmatrix} \end{matrix}$$

UPDATE D_z from D_i c

$$\begin{aligned} D_z(w) &= \min \{ c(z, x) + D_x(w), c(z, y) + D_y(w) \} \\ &= \min \{ 7 + 1, 2 + 3 \} = 5 \\ D_z(x) &= \min \{ c(z, x) + D_x(x), c(z, y) + D_y(x) \} \\ &= \min \{ 7 + 0, 2 + \infty \} = 7 \\ D_z(y) &= \min \{ c(z, x) + D_x(y), c(z, y) + D_y(y) \} \\ &= \min \{ 7 + \infty, 2 + 0 \} = 2 \\ D_z(z) &= 0 \end{aligned}$$

New D_z

$$\begin{bmatrix} 5 & 7 & 2 & 0 \\ w & x & y & z \end{bmatrix}$$

C

$$C = \begin{matrix} & \begin{matrix} w & x & y & z \end{matrix} \\ \begin{matrix} w \\ x \\ y \\ z \end{matrix} & \begin{bmatrix} 0 & 1 & 3 & \infty \\ 1 & 0 & \infty & 7 \\ 3 & \infty & 0 & 2 \\ \infty & 7 & 2 & 0 \end{bmatrix} \end{matrix}$$

Initial Vectors

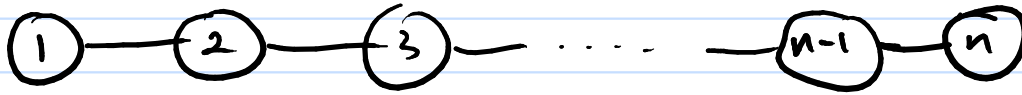
$$\begin{matrix} & \begin{matrix} w & x & y & z \end{matrix} \\ \begin{matrix} D_w \\ D_x \\ D_y \\ D_z \end{matrix} & \begin{bmatrix} 0 & 1 & 3 & \infty \\ 1 & 0 & \infty & 7 \\ 3 & \infty & 0 & 2 \\ \infty & 7 & 2 & 0 \end{bmatrix} \end{matrix}$$

* First D matrix, after neighbor updates, for w, x, y ?

* Updated distance vector for w, x, y ?

* How many rounds of updates until optimal D is found at ALL NODES?

Consider the Graph



• How many rounds until **1** discovers **n** / **n** discovers **1**?